

FEE 411: CONTROL SYSTEMS A (45 HOURS)

Prerequisites: FEE 222, FEE 372

Purpose of the Course

To introduce control engineering concepts and example control systems and provide analytical skills for time- and frequency-domain analysis of control systems.

Expected Learning Outcomes

At the end of this course, the learner should be able to:

- 1) Describe the working principles of a given control system
- 2) Analyse a control system solution in the time- and frequency-domains
- 3) Design a control system to control operations of a simple application
- 4) Analyse the stability of a control system and design a strategy for its enhancement

Course content

Introduction: Examples of control systems. Open and closed-loop classification. Control systems components and devices. Servomechanism, Stepper motors, Hydraulic systems, Pneumatic systems, Valves, cylinders, and rotary actuators.

Time Domain Linear Systems Analysis: Inputs, plant and outputs, Block diagrams, Mathematical models. The characteristic equation. Linear approximation to nonlinear systems. Analysis of linear (causal) systems as exemplified by first and second-order systems. Steady-state and transient responses. Continuous and discrete time-invariant systems. Delay time.

Transfer Functions: Laplace transform solution to a linear constant-coefficient differential equation. Transfer functions as developed from system differential equations. Definitions and properties in the s-plane. Block diagrams and canonical feedback. Derivation of transfer functions from the reduction of elementary block diagrams. Construction of signal flow diagrams. Block diagram reduction using signal flow diagrams.

S-Domain Analysis: Characteristic equation, Pole-zero map, System stability, Routh-Hurwitz stability criterion, Root locus. Standard inputs. Steady-state errors. System design and simulation.

Frequency Domain Linear Systems Analysis: Transfer function Frequency Response, Bandwidth, Nyquist Analysis: Complex functions of a complex variable. Singularity of a function. Closed contour and encirclement. Mapping properties of a complex function in the s-plane. Polar plots and their properties. Nyquist path, Nyquist plot, Nyquist stability criterion, Relative stability, Bode plot, Gain margin, Phase margin, Relative stability, Resonant peak, Resonant frequency, Nichols chart, Nichols plot.

Mode of Delivery

Lectures, tutorials, group-based learning, class exercises, assignments (face-to-face or ODEL-based).

2-hour lectures and 1-hour tutorial per week.

Instructional Materials and/or Equipment

Laptops, Tablets, LCD Projectors, Internet, Whiteboard and Stationery.

Course Assessment

Continuous Assessment Tests: 30%

Main Examination: 70%

Core Reading materials

- 1) Nagrath, I. J., & Gopal, M. (2009). *Control Systems Engineering* (5th Edition). New Age Publishers.
- 2) Norman, S. N. (2015). *Control Systems Engineering* (7th Edition). John Wiley and Sons Inc.
- 3) Petkov, H. P., Slavov, T. N., & Krlev, J. K. (2018). *Design of Embedded Robust Control Systems Using MATLAB®/SIMULINK®* (1st Edition). Institute of Engineering and Technology (IET).

Recommended Reference Materials

- 1) Distefano, J. (1990). *Theory and Problems of Feedback and Control Systems* (2nd Edition). Schaums/McGraw Hill.
- 2) Dorf, R., & Bishop, R. (2017). *Modern Control Systems* (13th Edition). Pearson.